TRINADH KUMAR REDDI

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Aspiring AI/ML Engineer with hands-on experience building RAG systems, data analysis pipelines, and full-stack applications using Python, LangGraph, and modern data tooling.

EDUCATION

B.Tech, Artificial Intelligence and Machine Learning — CGPA: 7.86
Bonam Venkata Chalamayya Engineering College, Odalarevu | 2021 - 2025

Intermediate (MPC) — 81.4% | Sri Chaitanya Jr College, Amalapuram | 2019 - 2021
Class X — 93% | G R Mariappan Sir C V Raman (EM) P S, Amalapuram | 2018 - 2019

PROJECTS

RAG Customer Support Assistant — LangGraph, ChromaDB & Streamlit

- Built a production-style Retrieval-Augmented Generation system for customer support using LangGraph, ChromaDB, and Streamlit as part of an internship project at Innomatics Research Labs.
- Implemented a PDF ingestion pipeline that chunks documents, generates embeddings with the BAAI/bge-small-en-v1.5 model, and stores them in ChromaDB for semantic retrieval.
- Designed a LangGraph workflow with intent detection and conditional routing that generates answers via Groq/LLaMA3 or OpenAI based on retrieved context and similarity confidence scores.
- Added a Human-in-the-Loop escalation flow that routes low-confidence, uncertain, or complex queries to a human agent panel, with responses integrated back into the chat as verified answers.
- Delivered a Streamlit UI supporting multi-PDF uploads, streaming responses, source citations, an explainable retrieval viewer, feedback buttons, and an analytics dashboard.

Risk-Return Evaluation of Indian Mutual Funds — Web Scraping & Data Analysis

- Scraped financial data from online sources using Python, BeautifulSoup, and Requests.
- Analyzed mutual fund performance metrics including Average Return, Volatility, Sharpe Ratio, AUM, and TER using Pandas and NumPy.
- Performed exploratory data analysis and visualizations using Matplotlib and Plotly to identify optimal investment categories.
- Generated insights showing ETFs and Index Funds as strong risk-adjusted investment options.

Brain Stroke Prediction — Exploratory Data Analysis

- Performed end-to-end data cleaning and EDA using Python, Pandas, and Matplotlib, applying IQR for outliers, creating pivot tables and visualizations, and uncovering key relationships between age, hypertension, heart disease, glucose level, smoking habits, and stroke across 5000+ patient records.

Bank Account Simulation System

- Built a console-based banking system using Python, implementing functions, loops, and conditional statements, with a Streamlit interface to simulate account creation, deposits, withdrawals, and balance checks.

Handwritten Character Recognition

- Built a character recognition model to classify handwritten alphabets and digits from image data, with the design extensible toward recognizing full words and sentences.

SKILLS

Programming Languages:	Python, SQL
Data Analysis & Visualization:	Pandas, NumPy, Matplotlib, Plotly, Power BI
Web Scraping & Data Collection:	BeautifulSoup, Requests, Regular Expressions (Regex)
Machine Learning:	Scikit-learn, Deep Learning Basics, NLP, RAG, LangGraph
Databases:	MySQL, RDBMS, SQL Queries, Joins
Tools:	Jupyter Notebook, MySQL, VS Code, Git & GitHub, Streamlit, Power BI, MS Excel
Soft Skills:	Teamwork, Critical Thinking, Communication, Problem Solving

HOBBIES & INTERESTS

Building end-to-end AI/ML and full-stack side projects
Exploring new developer tools and workflows (e.g. GitHub Copilot)
Competitive coding and problem solving
Reading about emerging AI research and applications
Playing chess
Cricket
Traveling

COURSES AND CERTIFICATES

Microsoft Certified: Azure AI Fundamentals — Microsoft, June 19, 2024
Machine Learning with Python (ML0101EN) — Cognitive Class (IBM), February 12, 2024
Python — GUVI, August 6, 2023